DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 12

✈ Chapter 12: Ground Radar

ATC Radar Systems

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1. Introduction

ATC uses ground radars extensively to serve a large number of requirements. Both primary and secondary radar techniques are used. Primary radar detects all aircraft (including those without transponders) and may incorporate Moving Target Indication (MTI).

📡 ATC Ground Radar Systems

- Area Surveillance Radar (ASR) — long range en-route
- Terminal Area Surveillance Radar (TAR) — medium range TMA
- Aerodrome Surveillance Approach Radar — short range approach
- Precision Approach Radar (PAR) — talk-down
- Airport Surface Movement Radar (ASMR/ASMI) — surface movement

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2. Area Surveillance Radar (ASR)

Introduction

Air Traffic Control services use ground radars extensively to serve a large number of requirements and users. They employ both primary radar and secondary radar techniques. Primary Radar is used to detect aircraft not equipped with a Secondary Radar Transponder. It may incorporate Moving Target Indication (MTI). The services that can be offered by Air Traffic Controllers are Information, Surveillance or Guidance.

Primary radar systems used by ATC include:

- Area Surveillance Radar (ASR)
- Terminal Area Surveillance Radar (TAR)
- Aerodrome Surveillance Radar
- Precision Approach Radar (PAR)
- Airport Surface Movement Radar (ASMR)

Area Surveillance Radars (ASR)

These are long range radars (200 to 300 NM) used for airway surveillance to provide range and bearing of aircraft. (Additional information is provided by Secondary Surveillance Radar - SSR). *Figures 12.1* and *12.2* show the locations and coverage of the London ACC and Scottish ACC radars and *Figure 12.3* shows the UK Airways structure.



Courtesy of Airbus Industrie

Figure 12.1 Coverage of LACC radars



Courtesy of Airbus Industrie

Figure 12.2 Coverage of SACC radars

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Fig 12.1 & 12.2: Coverage of London ACC and Scottish ACC radars

ASR Characteristics

Parameter	Value
Range	**200–300 NM**
Wavelength	**10–50 cm**
Pulse length	**2–4 µs**
PRF	**300–400 pps**
Scan rate	**5–6 rpm**
Purpose	Airway surveillance — range and bearing of aircraft

Longer pulse length ensures good target illumination at long range. Low PRF ensures echo returns from max range before next pulse.

3. Terminal Surveillance Area Radar (TAR)

TAR Characteristics

Parameter	Value
Range	Up to **75 NM**
Wavelength	**10, 23 or 50 cm**
Pulse width	**1–3 µs**
Purpose	Traffic control in TMAs
UK radar separation	**3 NM** within 40 NM (or 60 NM approved) of radar head, below FL245

4. Aerodrome Surveillance Approach Radar

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Aerodrome Radar Characteristics

Parameter	Value
Range	Up to **25 NM**
Wavelength	**3 cm or 10 cm**
Pulse width	**0.5–1 µs**
Services provided	Radar vectoring to ILS; Surveillance Radar Approach (SRA); positional information/control in vicinity

5. Airport Surface Movement Radar (ASMI)

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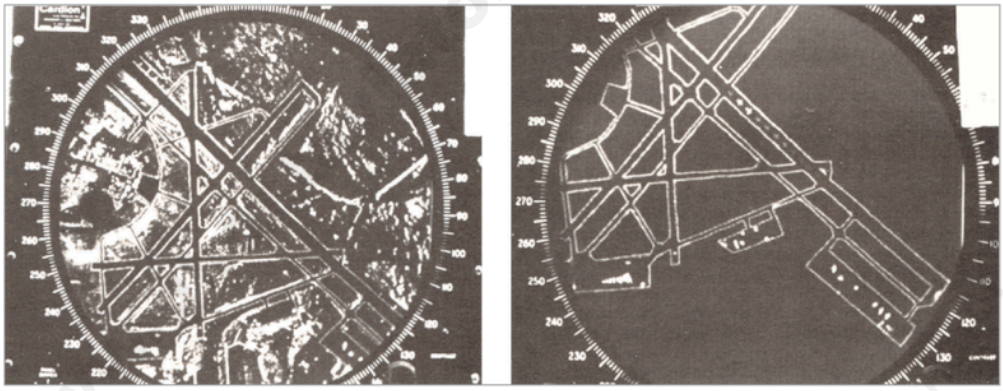


Figure 12.4

ASMI with fixed features Processed ASMI

Fig 12.4: ASMI display — with and without fixed features processed out

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⚠ ASMI Key Design Parameters

Parameter	Value
Frequency	**15–17 GHz (SHF)**
Wavelength	**1.76–2 cm**
Beamwidth	**0.2°–1°** (very narrow)
Scan rate	**60 rpm**
PRF	**4,000–20,000 pps**
Pulse width	**0.03 µs**
Range	**2.5–6 NM** in light precipitation
Purpose	Detailed display of aircraft/vehicles on runways and taxiways in all weather

⚡ ASMI Limitations

- High frequency (SHF) → increasingly attenuated by precipitation → range reduces in heavy rain
- EHF band NOT suitable for ASMI — too much attenuation in most precipitation types
- Processing can remove selected fixed features (buildings, taxiways) leaving only moving targets visible

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6. Radar Characteristics Summary

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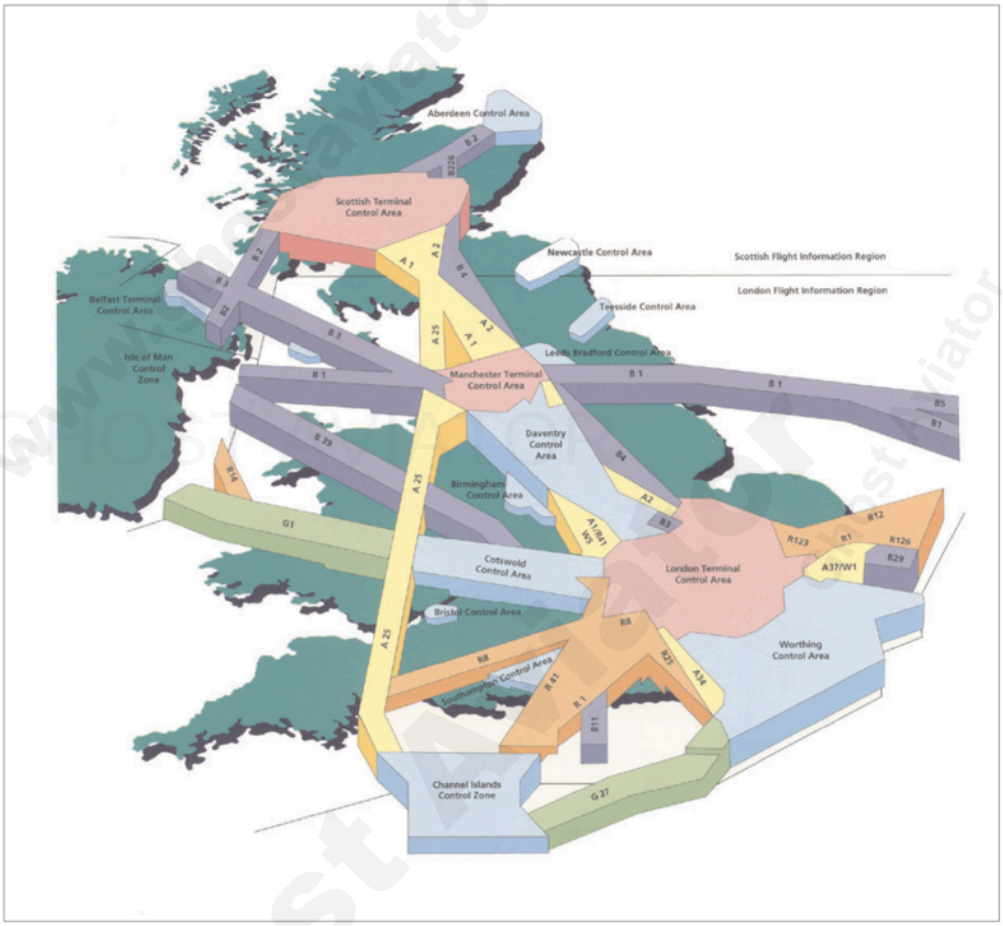


Figure 12.3 Airways in UK Airspace.

For the long range radars the wavelengths and pulse lengths are relatively long (10 to 50 cm and 2 to 4 μs respectively). The longer pulse length ensures that the target is illuminated for sufficient time to give a good return. The PRF and antenna rotation rate (scan rate) are low - 300 to 400 pps and 5 to 6 rpm respectively. This ensures that the next pulse is not transmitted until the first one has had sufficient time to return from the long range target.

Terminal Surveillance Area Radars

These are medium range radars, up to 75 NM, used for controlling traffic in TMAs. (Additional information is provided by Secondary Surveillance Radar - SSR).

Typical wavelengths are 10 cm, 23 cm and 50 cm with pulse widths 1 to 3 μs.

In the UK horizontal radar separation minima may be reduced to 3 NM (5.6 km) within 40 NM (or in certain circumstances 60 NM) of the radar head and below FL245 where the procedure has been officially approved.

Fig 12.3: UK airways structure — served by ground radars

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System	Range	Wavelength	PW	PRF	Scan Rate
ASR	200–300 NM	10–50 cm	2–4 µs	300–400 pps	5–6 rpm
TAR	Up to 75 NM	10, 23, 50 cm	1–3 µs	Medium	Medium
Aerodrome	Up to 25 NM	3 or 10 cm	0.5–1 µs	Higher	Higher
ASMI	2.5–6 NM	1.76–2 cm	0.03 µs	4,000–20,000 pps	60 rpm

7. Practice Questions

Q1. A primary radar has a PRF of 275 pps. Time interval between leading edges of successive pulses:

(a) 3.64 milliseconds

(b) 36.4 milliseconds

(c) 3.64 microseconds

(d) 36.4 microseconds

> **Answer: (a)**
> PRI = 1/275 s = 1,000,000/275 µs = 3636 µs = **3.64 milliseconds**.

Q2. Primary radar PRF = 450 pps. Maximum range (ignoring PW and flyback):

(a) 333 NM

(b) 180 NM

(c) 666 NM

(d) 360 NM

> **Answer: (b)**
> Max range = 81,000/450 = **180 NM**.

Q3. Frequency band and scan rate of Airfield Surface Movement radars:

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(a) SHF; 60 rpm

(b) SHF; 200 rpm

(c) EHF; 100 rpm

(d) EHF; 10 rpm

Answer: (a)
ASMI operates in the **SHF band (15–17 GHz)** at a scan rate of **60 rpm**.

Q4. Radar with scanner rotation 60 rpm, beamwidth ~0.5°, PRF 10,000 pps. This is:

(a) ASMI with theoretical range 8 NM

(b) Precision Approach Radar

(c) ASMI with theoretical range 16 NM

(d) High resolution Surveillance Approach Radar

Answer: (a)
All characteristics (60 rpm, narrow beam, high PRF, short PW) match an **ASMI**. Max range = 81,000/10,000 = 8.1 NM ≈ **8 NM**.

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